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Introduction

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1.1 Overview of Microwave and RF Engineering

Modern microwave and radio frequency (RF) engineering is an exciting and dynamic field, due in large part to the symbiosis between recent advances in modern electronic device technology and the current explosion in demand for voice, data, and video communication capacity. Prior to this revolution in communications, microwave technology was the nearly exclusive domain of the defense industry; the recent and dramatic increase in demand for communication systems for such applications as wireless paging, mobile telephony, broadcast video, and tethered as well as untethered computer networks is revolutionizing the industry. These systems are being employed across a broad range of environments including corporate offices, industrial and manufacturing facilities, and infrastructure for municipalities, as well as private homes. The diversity of applications and operational environments has led, through the accompanying high production volumes, to tremendous advances in cost-efficient manufacturing capabilities of microwave and RF products. This, in turn, has lowered the implementation cost of a host of new and cost-effective wireless as well as wired RF and microwave services. Inexpensive handheld GPS navigational aids, automotive collision-avoidance radar, and widely available broadband digital service access are among these. Microwave technology is naturally suited for these emerging applications in communications and sensing, since the high operational frequencies permit both large numbers of independent channels for the wide variety of uses envisioned as well as significant available bandwidth per channel for high speed communication.

Loosely speaking, the fields of microwave and RF engineering together encompass the design and implementation of electronic systems utilizing frequencies in the electromagnetic spectrum from approximately 300 kHz to over 100 GHz. The term "RF" engineering is typically used to refer to circuits and systems having frequencies in the range from approximately 300 kHz at the low end to between 300 MHz and 1 GHz at the high end. The term "microwave engineering", meanwhile, is used rather loosely to refer to design and implementation of electronic systems with operating frequencies in the range of from 300 MHz to 1 GHz on the low end to upward of 100 GHz. Figure 1.1 illustrates schematically the electromagnetic spectrum from audio frequencies through cosmic rays. The RF frequency spectrum covers the medium frequency (MF), high frequency (HF), and very high frequency (VHF) bands, while the microwave portion of the electromagnetic spectrum extends from the upper edge of the VHF frequency range to just below the THz radiation and far-infrared optical frequencies (approximately 0.3 THz and above). The wavelength of free-space radiation for frequencies in the RF frequency range is from approximately 1 m (at 300 MHz) to 1 km (at 300 kHz), while those of the microwave range extend from 1 m to the vicinity of 1 mm (corresponding to 300 GHz) and below.

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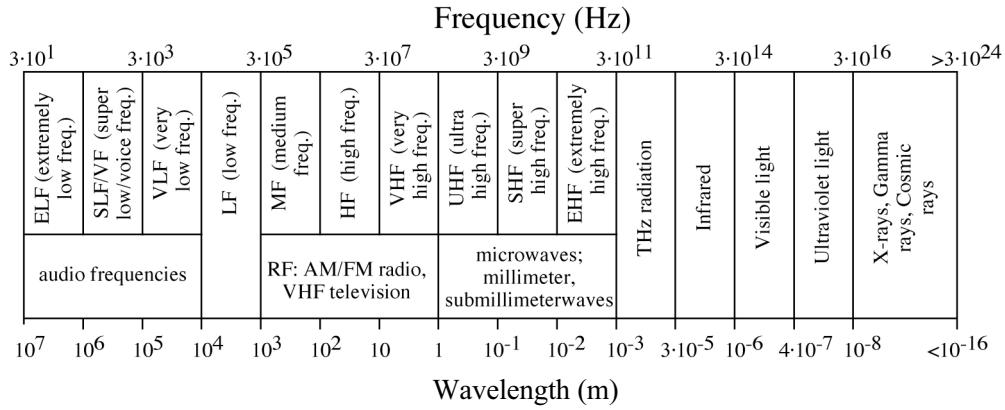

FIGURE 1.1 Electromagnetic frequency spectrum and associated wavelengths.

The boundary between "RF" and "microwave" design is both somewhat indistinct as well as one that is continually shifting as device technologies and design methodologies advance. This is due to implicit connotations that have come to be associated with the terms "RF" and "microwave" as the field has developed. In addition to the distinction based on the frequency ranges discussed previously, the fields of RF and microwave engineering are also often distinguished by other system features as well. For example, the particular active and passive devices used, the system applications pursued, and the design techniques and overall mindset employed all play a role in defining the fields of microwave and RF engineering. These connotations within the popular meaning of microwave and RF engineering arise fundamentally from the frequencies employed, but often not in a direct or absolute sense. For example, because advances in technology often considerably improve the high frequency performance of electronic devices, the correlation between particular types of electronic devices and particular frequency ranges is a fluid one. Similarly, new system concepts and designs are reshaping the applications landscape, with mass market designs utilizing ever higher frequencies rapidly breaking down conventional notions of microwave-frequency systems as serving "niche" markets.

The most fundamental characteristic that distinguishes RF engineering from microwave engineering is directly related to the frequency (and thus the wavelength) of the electronic signals being processed. For low-frequency and RF circuits (with a few special exceptions such as antennae), the signal wavelength is much larger than the size of the electronic system and circuit components. In contrast, for a microwave system the sizes of typical electronic components are often comparable (i.e., within approximately 1 order of magnitude) to the signal wavelength. This gives rise to a reasonable working definition of the two design areas based on the underlying approximations used in design. Since in conventional RF design, the circuit components and interconnections are generally small compared to a wavelength, they can be modeled as lumped elements for which Kirchoff's voltage and current laws apply at every instant in time. Parasitic inductances and capacitances are incorporated to accurately model the frequency dependencies, but these quantities can, to good approximation, be treated as lumped elements. For microwave frequencies, however, the finite propagation velocity of electromagnetic waves can no longer be neglected. For these frequencies, the time delay associated with signal propagation from one end of a component to the other is an appreciable fraction of the signal period, and thus lumped-element descriptions are no longer adequate to describe the electrical behavior. A distributed-element model is required to accurately capture the electrical behavior, in contrast to RF models. The time delay associated with finite wave propagation velocity that gives rise to the distributed circuit effects is a distinguishing feature of the mindset of microwave engineering. An alternative viewpoint is based on the observation that microwave engineering lies in a "middle ground" between traditional low-frequency electronics and optics, as shown in Fig. 1.1. As a consequence of RF, microwaves, and optics simply being different regimes of the same electromagnetic phenomena, there is a gradual transition between these regimes. The continuity of these regimes

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results in constant reevaluation of the appropriate design strategies and trade-offs as device and circuit technology advances. For example, miniaturization of active and passive components often increases the frequencies at which lumped-element circuit models are sufficiently accurate, since by reducing component dimensions, the time delay for propagation through a component is proportionally reduced. As a consequence, lumped-element components at "microwave" frequencies are becoming increasingly common in systems previously based on distributed elements due to significant advances in miniaturization, even though the operational frequencies remain unchanged. Component and circuit miniaturization also leads to tighter packing of interconnects and components, potentially introducing new parasitic coupling and distributed-element effects into circuits that could previously be treated using lumped-element RF models.

The comparable scales of components and signal wavelengths has other implications for the designer as well, since neither the ray-tracing approach from optics nor the lumped-element approach from RF circuit design are valid in this middle ground. In this regard, microwave engineering can also be considered to be "applied electromagnetic engineering," as the design of guided-wave structures such as waveguides and transmission lines, transitions between different types of transmission lines, and antennae all require analysis and control of the underlying electromagnetic fields. The design of these guided-wave structures is treated in detail in Section 9.3.

The distinction between RF and microwave engineering is further blurred by the trend of increasing commercialization and consumerization of systems using what have been traditionally considered to be microwave frequencies. Traditional microwave engineering, with its historically military applications, has long been focused on delivering performance at any cost. As a consequence, special-purpose devices intended solely for use in high performance microwave systems and often with somewhat narrow ranges of applicability were developed to achieve the required performance. With continuing advances in silicon microelectronics, including SiGe heterojunction bipolar transistors (HBTs) and conventional scaled CMOS, microwave frequency systems can now be reasonably implemented using the same devices as conventional low-frequency baseband electronics. In addition, the commercialization of low-cost III-V compound semiconductor electronics, including ion-implanted metal-semiconductor field-effect transistors (MESFETs), pseudomorphic high electron mobility transistors (PHEMTs), and III-V HBTs, has dramatically decreased the cost of including these elements in high-volume consumer systems. This convergence, with silicon microelectronics moving ever higher in frequency into the microwave spectrum from the low-frequency side and compound semiconductors declining in price for the middle of the frequency range, blurs the distinction between "microwave" and "RF" engineering, since "microwave" functions can now be realized with "mainstream" low-cost electronics. This is accompanied by a shift from physically large, low-integration-level hybrid implementations to highly integrated solutions based on monolithic microwave integrated circuits (MMICs). This shift has a dramatic effect not only on the design of systems and components, but also on the manufacturing technology and economics of production and implementation as well. A more complete discussion of the active device and integration technologies that make this progression possible is included in Chapter 7.

Aside from these defining characteristics of RF and microwave systems, a number of physical effects that are negligible at lower frequencies become increasingly important at high frequencies. Two of these effects are the skin effect and radiation losses. The skin effect is caused by the finite penetration depth of an electromagnetic field into conducting material. This effect is a function of frequency; the depth of penetration is given by $\delta_s = \frac{1}{\sqrt{\omega f \mu \sigma}}$, where μ is the permeability, f is the frequency, and σ is the conductivity of the material. As the expression indicates, δ_s decreases with increasing frequency, and so the electromagnetic fields are confined to regions increasingly near the surface as the frequency increases. This results in the microwave currents flowing exclusively along the surface of the conductor, significantly increasing the effective resistance (and thus the loss) of metallic interconnects. Further discussion of this topic can be found in Sections 9.2 and 9.6.1. Radiation losses also become increasingly important as the signal wavelengths approach the component and interconnect dimensions. For conductors and other components of comparable size to the signal wavelengths, standing waves caused by reflection of the electromagnetic waves from the boundaries of the component can greatly enhance the radiation of

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electromagnetic energy. These standing waves can be easily established either intentionally (in the case of antennae and resonant structures) or unintentionally (in the case of abrupt transitions, poor circuit layout, or other imperfections). Careful attention to transmission line geometry, placement relative to other components, transmission lines, and ground planes, as well as circuit packaging is essential for avoiding excessive signal attenuation and unintended coupling due to radiative effects.

A further distinction in the practice of RF and microwave engineering from conventional electronics is the methodology of testing. Due to the high frequencies involved, the capacitance and standing-wave effects associated with test cables and the parasitic capacitance of conventional test probes make the use of conventional low-frequency circuit characterization techniques impractical. Although advanced measurement techniques such as electro-optic sampling can sometimes be employed to circumvent these difficulties, in general the loading effect of measurement equipment poses significant measurement challenges for debugging and analyzing circuit performance, especially for nodes at the interior of the circuit under test. In addition, for circuits employing dielectric or hollow guided-wave structures, voltage and current often cannot be uniquely defined. Even for structures in which voltage and current are well-defined, practical difficulties associated with accurately measuring such high-frequency signals make this difficult. Furthermore, since a DC-coupled time-domain measurement of a microwave signal would have an extremely wide noise bandwidth, the sensitivity of the measurement would be inadequate. For these reasons, components and low-level subsystems are characterized using specialized techniques. These approaches are treated in detail in Chapter 4.

1.2 Frequency Band Definitions

The field of microwave and RF engineering is driven by applications — originally for military purposes such as radar and, more recently, for commercial, scientific, and consumer applications. As a consequence of this diverse applications base, microwave terminology and frequency band designations are not entirely standardized, with various standards bodies, corporations, and other interested parties all contributing to the collective terminology of microwave engineering. Figure 1.2 shows graphically some of the most common frequency band designations, with their approximate upper and lower bounds. As can be seen, some care must be exercised in the use of the "standard" letter designations; substantial differences in the definitions of these bands exist in the literature and in practice. Diagonal hashing at the ends of the frequency bands in Fig. 1.2 indicates variations in the definitions by different groups and authors; dark regions in the bars indicate frequencies for which there appears to be widespread agreement in the literature. The double-ended arrows appearing above some of the bands indicate the IEEE definitions for these bands. Two distinct definitions of K-band are in use; the first of these defines the band as the range from 18 GHz to approximately 26.5 GHz, while the other definition extends from 10.9 to 36 GHz. Both of these definitions are illustrated in Fig. 1.2. Similarly, L-band has two overlapping frequency range definitions; this gives rise to the large "variation" regions shown in Fig. 1.2. In addition, some care must be taken with these letter designations, as the IEEE and U.S. military specifications both define an L-band, but with very different frequencies. The IEEE L-band resides at the low end of the microwave spectrum, while the military definition of L-band is from 40 to 60 GHz. The IEEE designations (L–W) are presently used widely in practice and the technical literature, with the newer U.S. military designations (A–N) having not yet gained widespread popularity outside of the military community.

1.3 Applications

The field of microwave engineering is currently experiencing a radical transformation. Historically, the field has been driven by applications requiring the utmost in performance with little concern for cost or manufacturability. These systems have been primarily for military applications, where performance at nearly any cost could be justified. The current transformation of the field involves a dramatic shift from defense applications to those driven by the commercial and consumer sector, with an attendant shift in

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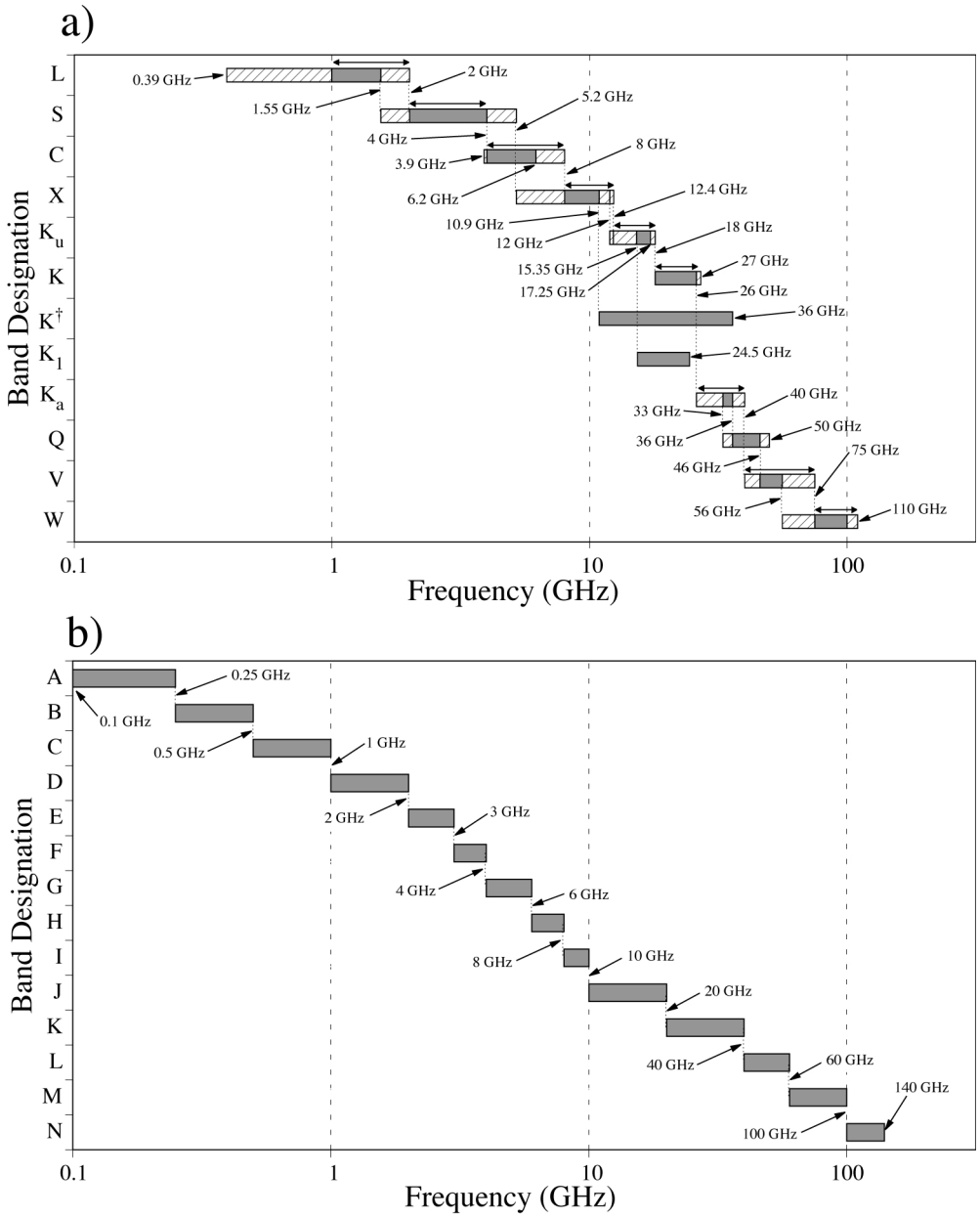

FIGURE 1.2 Microwave and RF frequency band designations.[1-5] (a) Industrial and IEEE designations. Diagonal hashing indicates variation in the definitions found in literature; dark regions in the bars indicate frequencies for which there is widespread agreement. Double-ended arrows appearing on bands indicate the current IEEE definitions for these bands where they exist, and K† denotes an alternative definition for K-band found in Reference 5. (b) U.S. military frequency band designations.[1-3]

focus from design for performance to design for manufacturability. This transformation also entails a shift from small production volumes to mass production for the commercial market, and from a focus on performance without regard to cost to a focus on minimum cost while maintaining acceptable performance. For wireless applications, an additional shift from broadband systems to systems having very tightly regulated spectral characteristics also accompanies this transformation.

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For many years the driving application of microwave technology was military radar. The small wavelength of microwaves permits the realization of narrowly focused beams to be achieved with antennas small enough to be practically steered, resulting in adequate resolution of target location. Long-distance terrestrial communications for telephony as well as satellite uplink and downlink for voice and video were among the first commercially viable applications of microwave technology. These commercial communications applications were successful because microwave-frequency carriers (f_c) offer the possibility of very wide absolute signal bandwidths (Δf) while still maintaining relatively narrow fractional bandwidths (i.e., $\Delta f/f_c$). This allows many more voice and data channels to be accommodated than would be possible with lower-frequency carriers or baseband transmission.

Among the current host of emerging applications, many are based largely on this same principle, namely, the need to transmit more and more data at high speed, and thus the need for many communication channels with wide bandwidths. Wireless communication of voice and data, both to and from individual users as well as from users and central offices in aggregate, wired communication including coaxial cable systems for video distribution and broadband digital access, fiber-optic communication systems for long- and short-haul telecommunication, and hybrid systems such as hybrid fiber-coax systems are all poised to take advantage of the wide bandwidths and consequently high data carrying capacity of microwave-frequency electronic systems. In addition to the explosion in both diversity and capability of microwave-frequency communication systems, radar systems continue to be of importance with non-military and non-navigational applications including radar systems for automotive collision avoidance and weather and atmospheric sensing becoming increasingly widespread.

In addition to these "traditional" microwave applications, other fields of electronics are increasingly encroaching into the microwave frequency range. Examples include wired data networks based on coaxial cable or twisted-pair transmission lines with bit rates of over 1 Gb/s, fiber-optic communication systems with data rates well in excess of 10 Gb/s, and inexpensive personal computers and other digital systems with clock rates of over 1 GHz. The continuing advances in the speed and capability of conventional microelectronics is pushing traditional circuit design ever further into the microwave frequency regime. These trends promise to both invigorate and reshape the field of microwave engineering in new and exciting ways.

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